Jiho Hwang

Undergraduate student in Chemistry · Seoul National University

jhhwang16@snu.ac.kr · GitHub · Website

Education

Mar 2026–present

B.S. in Chemistry

Seoul National University

In progress

Mar 2023–Jan 2026

High School Diploma

Gyeonggi Science High School

GPA: 4.19 / 4.30

Research Experience

Apr 2025–Jan 2026

Creative R&E

Gyeonggi Science High School · KOFAC

- Computational investigation of organic reaction mechanisms and thermochemistry.
- Activation/strain analysis, energy decomposition analysis, and QTAIM.

Mar–Nov 2024

Advanced R&E in Chemistry

Gyeonggi Science High School

- Investigated polarity conversion using implicit-solvation models.
- Studied conformational fluctuations in a sterically constrained molecule.
- Performed excited-state calculations using TD-DFT and EOM-CC methods.

Honors and Awards

Feb 2026

Silver Prize, 32nd Samsung Humantech Paper Awards

High School Chemistry Division · Second place

Jan 2026

Talent Award of Future Chemist

The Korean Union of Chemical Science and Technology Societies

Jan 2026

Certificate of Excellence in Chemistry

Gyeonggi Science High School

Feb 2025

National-team candidate, 57th International Chemistry Olympiad

Korean Chemistry Olympiad Winter School · Top eight

Feb 2025

Gold Prize, 31st Samsung Humantech Paper Awards

High School Chemistry Division · First place

Conference Presentations

Uncovering origins of the reaction driving force in an Aromatic Ring-Opening/Ring-Closing Metathesis cascade

Jiho Hwang, Seungchan Hong, Doyoung Kim, and Jaehoon Min

- **ACS Spring Meeting 2026**, March 2026. ORGN poster presentation; selected for Sci-Mix.
- **Korean Chemical Society Fall Conference**, September 2025. Future Chemist Research Presentation; poster.

Activities

2023–2025	**President, Alchemist** Experimental Chemistry Club, Gyeonggi Science High School
2024	**Vice President, TeXperT and GSHS TeX Society** Gyeonggi Science High School

Professional Affiliations

- American Chemical Society · Undergraduate member
- ACS Division of Organic Chemistry (ORGN) · Undergraduate member

Skills

Chemistry software	Gaussian, ORCA, CFOUR, Multiwfn
Programming	Python, C++
Languages	Korean (native), English, Chinese (HSK Level 4)

Research Interests

Organic synthesis. Organic and asymmetric synthesis, natural-product synthesis, organometallic catalysis, and synthetic methodology development.

Reaction mechanisms and thermochemistry. Reaction driving forces and the analysis of molecular reactivity using activation/strain models, energy decomposition, and QTAIM.

Computational chemistry. Density functional theory, coupled-cluster methods, conformational analysis, and excited-state calculations.